

DESIGN AND IMPLEMENTATION OF KAL-READ & LEARN- AN AI-POWERED ANDROID APPLICATION FOR MOTIVATING READING HABITS THROUGH GAMIFIED QUIZZES

A Mini Project Report Submitted to The Bharathiar University in Partial Fulfilment of the
Requirements for The Award of the Degree Of

MASTER OF COMPUTER APPLICATIONS

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NOVEMBER - 2025

CERTIFICATE

This is to certify that this mini-project work entitled **DESIGN AND IMPLEMENTATION OF KAL-READ & LEARN-AN AI-POWERED ANDROID APPLICATION FOR MOTIVATING READING HABITS THROUGH GAMIFIED QUIZZES.**, was submitted to the Department of Computer Applications, Bharathiar University in partial fulfilment of the requirements for the award of the degree of **MASTER OF COMPUTER APPLICATIONS**, is a record of original work done by **ABHIMANYU T (24CSEA001)** during his period of study in the Department of Computer Applications, Bharathiar University, Coimbatore, under my supervision and guidance, and this project work has not formed the basis for the award of any Degree/ Diploma/Associateship/ Fellowship or similar title to any candidate of any University.

Place: Coimbatore

Date:

Project Guide

Head of the Department

Submitted for the University Viva-Voce Examination held on _____

Internal Examiner

External Examiner

DECLARATION

I hereby declare that this mini-project work titled, **DESIGN AND IMPLEMENTATION OF KAL-READ & LEARN-AN AI-POWERED ANDROID APPLICATION FOR MOTIVATING READING HABITS THROUGH GAMIFIED QUIZZES**, submitted to Department of Computer Applications, Bharathiar University, is a record of original work done by **ABHIMANYU T (24CSEA001)**, under the supervision and guidance of **Dr. T. AMUDHA, M.C.A., Ph.D.**, Professor, Department of Computer Applications, Bharathiar University, and that this project work has not formed the basis for the award of any Degree/ Diploma/ Associateship/ Fellowship or similar title to any candidate of any University.

Place: Coimbatore

Signature of the Candidate

Date:

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COUNTERSIGNED BY

ACKNOWLEDGEMENT

I have taken efforts in this project. However, it would not have been possible without the kind support and help of many individuals and organizations. I would like to extend my sincere thanks to all of them. I express my sincere gratitude to our **Dr. M. PUNITHAVALLI, M.Sc., M.Phil., Ph.D.,** Professor & Head, Department of Computer Application, Bharathiar University, Coimbatore, thanks to department faculty members and guest faculties supported to acquire knowledge from various inputs. I am forever indebted to this Mini project guide **Dr. T. AMUDHA, M.C.A., M.Phil., Ph.D.,** Professor, Department of Computer Applications, Bharathiar University for providing valuable suggestions and always been enthusiastically guiding us throughout the project.

Finally, I also extend my special thanks to my family, friends, who have kindly provided the necessary support for the successful completion of the mini-project and their moral support.

ABSTRACT

The KAL – Read & Learn application is an Android-based intelligent reading and learning platform designed to motivate users to build consistent reading habits through gamification, AI-generated quizzes, and a virtual reward ecosystem. The system integrates Firebase Authentication, Firestore, and Storage to provide secure user management, real-time data synchronization, and seamless access to open-source digital books. After reading a specified number of pages, the application triggers a Node.js middleware that extracts book content and communicates with Google's Gemini AI to generate context-aware quiz questions, enabling users to test comprehension instantly. Correct answers reward users with in-app coins, which can be used to unlock new books or publish their own creative writings such as poems and stories. The platform further supports authors by allowing them to advertise real-world books directly to a reader-rich audience. Built using Kotlin and Jetpack Compose, the app offers a modern, responsive UI and ensures smooth performance across Android devices. Overall, KAL provides a unified ecosystem that promotes reading, learning, creativity, and engagement through AI-driven evaluation and a coin-based progression system, making it a unique and interactive educational tool.

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1.INTRODUCTION

In today's digital era, people are gradually losing the habit of reading books due to the influence of short-form entertainment, social media distractions, and lack of motivation. Reading has become a passive activity rather than an engaging experience, leading to a decline in comprehension skills and overall interest in literature. Many existing book platforms only provide digital access to reading materials but fail to create a sense of reward, interaction, or learning progress. As a result, readers often abandon books midway, and creative expression among young readers remains untapped. At the same time, numerous aspiring writers and published authors struggle to find the right audience for their work. The KAL – Read & Learn app is designed to address this cultural and behavioral shift. It redefines reading as an interactive and rewarding experience through the concept of "Gamified Reading." The app transforms every reading session into a learning opportunity by integrating AI-generated quizzes that test comprehension and reward readers with App Coins for correct answers. These coins can be used to unlock premium books or publish their own creative content such as poems, stories, or thoughts. Moreover, KAL acts as a creative ecosystem where authors can advertise their real-world books to a relevant and active audience. Readers, in turn, become potential buyers and literary influencers. This creates a self-sustaining environment where learning, creativity, and marketing coexist. By merging gamification, AI-driven learning, and a creative social layer, KAL – Read & Learn not only encourages reading habits among users but also helps authors understand popular reader interests and trends. This insight allows authors to plan their next publications more effectively.

1.1 Problem Statement

In today's fast-paced digital world, people are losing interest in reading due to the rise of short-form entertainment and lack of motivation. Existing book platforms offer reading access but fail to make it interactive or rewarding. This results in poor retention, limited comprehension, and declining reading habits. At the same time, authors struggle to reach the right audience and promote their works effectively. There is no unified platform that connects readers and writers through engagement and learning. Hence, there is a need for an intelligent system that motivates reading, encourages creativity, and provides exposure for authors within a single ecosystem.

1.2 Objectives of the Study

The main objective of this study is to develop an interactive mobile application that motivates users to rebuild their reading habits through gamification and AI-driven engagement. The KAL – Read & Learn app aims to transform traditional reading into an active learning process by generating quizzes from book content using Gemini AI, rewarding users with App Coins for correct answers. These coins can be used to unlock premium books or publish creative thoughts like poems and stories, promoting continuous learning and self-expression. The app also provides a dedicated space for authors to advertise their real-world books to a relevant audience, creating a bridge between readers and writers. Additionally, it seeks to understand user reading interests, helping authors design better content. By combining AI, Firebase, and AdMob integration, the study focuses on building a sustainable reading ecosystem that enhances comprehension, creativity, and community interaction in the digital age.

1.3 Scope of the Project

The scope of this study focuses on designing and developing a smart, interactive, and user-friendly Android application that encourages reading habits through gamification and AI-based learning. The KAL – Read & Learn app integrates advanced technologies such as Gemini AI, Firebase, and Jetpack Compose to create a seamless reading and learning experience. It allows users to read open-source books, attend auto-generated quizzes, and earn App Coins, which can be used for unlocking premium content or publishing their own writings. The study extends to building a secure user authentication system with email and phone verification, ensuring personalized profiles for both users and authors. It also includes real-time data storage and quiz tracking using Firebase Firestore. For authors, the app provides a marketing platform to showcase and advertise their real-world books with cover pages, synopsis, and purchase links. The inclusion of Google AdMob and Engagement Puzzles enhances user engagement and sustains app revenue without monetary rewards. The project scope further covers UI/UX design, performance optimization, and data-driven insights to identify popular reading interests among users. Overall, the study aims to build a multifunctional digital platform that not only revives the reading culture but also connects readers, creators, and authors in a single ecosystem for educational and creative growth.

1.4 Proposed System

The proposed KAL – Read & Learn system aims to create an interactive and engaging reading ecosystem that transforms passive reading into an active learning experience. It integrates AI-based quiz generation using Gemini AI to test readers' comprehension after reading specific portions of a book. Users earn App Coins for correct answers, which can be used to unlock premium books or publish their own writings. The system also provides a common platform for authors to advertise their real-world books and reach an audience genuinely interested in reading. Thus, the proposed system encourages reading habits, promotes creativity, and connects readers and authors in one intelligent digital environment.

Features of the proposed System

1. User and Author Role Management:

The system supports two types of users Readers and Authors. Each role has a customized home screen while sharing common Explore and Profile sections, enabling easy navigation and engagement across both user groups.

2. Secure User Authentication:

Users can register and log in through Firebase Authentication using email verification, phone number, date of birth, and password, ensuring secure identity management and personalized experience.

3. AI-Based Quiz Generation:

After reading a minimum of ten pages, the system extracts text from uploaded PDFs using Node.js and generates intelligent quizzes through Gemini AI, testing user comprehension in real-time.

4. Reward and Coin System:

Users earn App Coins for every correct quiz answer. These coins serve as an in-app currency that can be used to unlock premium books or to publish personal writings such as poems, short stories, novels.

5. Content Publishing Platform:

Readers can express their creativity by posting their writings on the Explore page, where others can view, like, and engage with their content building a reading and writing community.

6. Author Book Promotion:

Authors can advertise their real-world books by uploading the book cover, synopsis, and buying links. They can also view user engagement metrics like likes and reach, reader interests.

7. Engagement Puzzle Integration:

To reduce waiting frustration during quiz generation, small puzzles or mini-games are displayed, maintaining user engagement and improving overall satisfaction.

8.AdMob Integration:

The app incorporates Google AdMob for displaying non-intrusive ads, generating platform revenue without affecting the user experience or introducing real-money transactions.

9.Data storage and Tracking:

All quiz results, user progress, earned coins, and creative posts are securely stored in Firebase Firestore, allowing users to view their reading history, wallet balance, and performance analytics.

10. Community and Feedback System:

The Explore section encourages social interaction where both users and authors can like, exchange ideas, creating a vibrant community that blends education, creativity, and engagement.

Advantages of the Proposed System:

The proposed system combines reading, learning, earning, and creativity into a single unified platform, offering several advantages over existing systems. It effectively motivates users to rebuild their reading habits through gamification and rewards, making learning enjoyable and measurable. The AI-driven quiz generation ensures active comprehension, while the coin-based economy encourages consistent participation. For authors, the system provides direct access to a validated, book-loving audience, enhancing the effectiveness of digital book marketing. Unlike other apps, KAL – Read & Learn creates a sustainable ecosystem where users grow as readers and creators, authors reach relevant audiences, and the platform maintains engagement through intelligent design and AI-powered interactivity. Additionally, the system promotes user creativity by allowing them to publish poems, stories, and thoughts using earned coins, thus encouraging literary expression. The use of Firebase ensures secure, scalable, and real-time data handling across all modules. Integration of AdMob and engagement puzzles maintains user interest during AI processing time. The application's adaptive structure ensures smooth functionality even on mid-range devices. Overall, KAL stands out as a next-generation learning and reading platform that blends technology, motivation, and creativity seamlessly.

2.LITERATURE REVIEW

The KAL – Read & Learn application is an innovative Android-based platform designed to revive the declining habit of book reading through gamification and AI integration. It transforms reading into an engaging and rewarding experience by generating AI-based quizzes from book content to test comprehension and enhance learning. Users earn App Coins for correct answers, which can be used to unlock premium books or publish their own creative writings. The app also serves as a digital space for authors to promote their real-world books to an active and relevant audience. Using technologies such as Kotlin, Jetpack Compose, Firebase, and Gemini AI, it ensures smooth functionality and intelligent content generation. The inclusion of AdMob ads and Engagement Puzzles sustains user interest and platform revenue. Overall, the app bridges the gap between readers and writers, creating a dynamic ecosystem for learning, creativity, and community engagement.

2.1 Existing Systems

1.Pratilipi

Pratilipi is one of India's largest online storytelling platforms that allows users to read and write stories in multiple Indian languages. It provides a digital space for both readers and authors to engage and share literary content easily.

Features:

It supports self-publishing for emerging authors, offering opportunities to reach wide audiences without intermediaries. The app includes personalized reading suggestions and multilingual story collections. It allows community engagement through comments and author-reader interactions. Pratilipi has introduced a creator monetization program where writers can earn based on engagement. It also provides audio versions of popular stories for better accessibility.

Limitations:

The platform currently lacks interactive learning features such as AI-based quizzes or gamified reading experiences. It does not include a structured reward or coin-based system to motivate continued reading. Authors can promote stories but have limited analytical tools for advertising performance. Readers do not receive any tangible rewards for participation. The app mainly focuses on storytelling rather than learning enhancement through quizzes or rewards.

2. Wattpad

Wattpad is a global online platform for reading and writing user-generated stories across various genres. It has a large Indian audience and has helped many authors publish and gain recognition.

Features:

Users can create, publish, and share their stories directly on the platform. Wattpad offers a social community where readers can comment, vote, and follow their favorite authors. It includes writing contests and paid story programs for popular creators. The app provides an immersive reading experience through serialized storytelling and community interactions. Wattpad also recommends personalized reading lists based on user interests.

Limitations:

Wattpad does not provide coin-based rewards or a structured earning system for general readers. The app lacks integrated quizzes or comprehension-based learning tools. Advertising and monetization opportunities are mostly limited to top-tier authors. It does not allow book unlocking through coins or gamified engagement. The platform's focus is on community storytelling rather than educational reading motivation.

3.BookTaco

BookTaco is an educational reading platform designed primarily for students, combining reading comprehension, quizzes, and rewards to encourage learning. It offers a structured approach where each book read is followed by quizzes to test understanding.

Features:

BookTaco includes a large library of books with interactive quizzes for each title. Students can earn virtual rewards and badges based on quiz performance. It supports teacher dashboards for tracking reading progress and comprehension. The system promotes reading through gamification and provides engagement analytics. Its integration of rewards and learning makes it effective for educational motivation.

Limitations:

The platform focuses on school-level content and lacks open publishing for independent authors. Readers cannot express creative thoughts or post original writing. It does not include advertising or promotional features for authors. The reward system is virtual and limited to classroom use, not public coins or unlockable content. BookTaco is designed for institutions, not individual readers or casual learning communities.

4. Cashzine

Cashzine is a mobile app where users can read articles, news, or stories and earn coins that can be exchanged for rewards. It is a “read-to-earn” platform aimed at increasing engagement.

Features:

It rewards users with coins for reading content daily, encouraging consistent usage. The app provides trending stories and personalized recommendations. It includes referral programs and bonus coins for specific activities. The interface is simple and gamified to attract a broad audience. Its reward model makes reading a more interactive and habitual experience.

Limitations:

Cashzine focuses mainly on short articles and news rather than structured books or educational content. It lacks depth in learning or quiz-based comprehension. Users cannot publish or advertise their own writing. The coins and reward mechanisms are often limited to ad-based incentives rather than meaningful engagement. It has no author-reader ecosystem or long-form content like novels or textbook.

2.2 Comparative Study

Table 1 compares popular reading platforms based on their monetization methods, engagement models, supported actions, languages, and level of gamification in relation

Table 1. Comparative Study

Application Name	Platform	Monetization Model	User Engagement Model	Supported Actions	Supported Languages	Level of Gamification
Pratilipi	Android, iOS, Web	Advertisements, Premium subscriptions, Author Monetization	Likes, Comments, Follows, Reading History	Read stories, Write stories, Listen to audiobooks, Follow authors	Multiple Indian languages (Tamil, Hindi, Telugu, Malayalam, Kannada, Bengali, etc.)	Very Low (No quizzes or rewards)
Wattpad	Android, iOS, Web	Ads, Premium subscription (Wattpad Premium), Paid Stories for writers	Votes, Comments, Reading lists, Following authors	Read, Write, Publish, Comment, Vote	English + Multiple global languages	Low (Social engagement only)
BookTaco	Web	Subscription (schools/teachers), Classroom licenses	Quiz points, Progress tracking, Badges (for students)	Read books (linked ISBN list), Take quizzes, Track performance	English	Medium (Quiz-based gamification for students)
Cashzine	Android, iOS	Ads, Coin-to-reward system (non-cash), Referral bonuses	Coin rewards, Daily check-ins, Reading tasks	Read articles, Earn coins, Daily tasks	English, Indonesian, Filipino, etc.	Medium (Coin-based engagement, but not educational)

3.SYSTEM ANALYSIS

System analysis is the process of understanding user requirements, defining system functionalities, and evaluating the practicality of the proposed solution. It forms the foundation for designing an efficient and user-centered software system

In this chapter, the system requirements of KAL – Read & Learn, an AI-powered reading motivation and learning platform, are analyzed. It includes the requirement analysis, feasibility study, and problem formulation, ensuring that the system is technically viable, operationally efficient, and fulfills its goal of transforming passive reading into an active, rewarding, and community-driven experience.

3.1 Requirement Analysis

Requirement analysis involves gathering, studying, and documenting what functionalities the proposed system must provide. This step ensures that the software meets user expectations and performs effectively under real-world conditions.

3.1.1 Functional Requirements

Functional requirements define what the system should do — the core features, behaviors, and interactions expected from the KAL – Read & Learn application. The Table 2 below lists and explains the key functional requirements of the system, describing how each feature contributes to user engagement, learning, and platform functionality.

3.1.2 Non-Functional Requirements

Non-functional requirements define the quality attributes that describe how the system performs rather than what it does. The table below lists the essential non-functional requirements of the KAL – Read & Learn application, ensuring system efficiency, reliability, and user satisfaction.

Table 2 outlines the main functional components of the KAL application, detailing their purpose and role in fulfilling the system’s objectives.

Table 2. Functional Requirements

S.No	Requirement	Description
1	User Authentication	Users can sign up, log in, and manage their profiles securely using Firebase Authentication. Registration requires name, email verification, phone number, date of birth, and password.
2	Role-Based Access	The system supports two distinct roles — User and Author. Users can read books and earn coins, while authors can advertise and promote real-world books.
3	Book Reading Module	Users can access open-source books stored in Firebase Storage. The reading progress is tracked to determine when a user becomes eligible for quizzes.
4	AI-Based Quiz Generation	After reading a minimum of ten pages, the Node.js backend extracts PDF text and uses Gemini AI to generate quiz questions, ensuring content relevance.
5	Quiz Evaluation & Coin Reward System	User answers are evaluated automatically. Each correct answer earns App Coins, stored securely in Firestore. Coins are used to unlock premium content or publish writings.
6	Creative Publishing Module	Users can spend coins to post poems, stories, or novels publicly on the Explore screen, where others can like and engage with their content.
7	Author Advertisement Module	Authors can upload their book covers, synopses, and purchase links to advertise their real-world publications to the reader community.
8	AdMob Integration	The app integrates Google AdMob to display relevant ads, ensuring monetization without disrupting user experience.
9	Engagement Puzzle	During AI quiz generation, users can play small puzzles to stay engaged and reduce waiting frustration.
10	User Data Storage & Analytics	All user progress, quiz results, and coin transactions are stored in Firebase Firestore.

Table 3 highlights the performance, security, scalability, and usability factors that govern how effectively the KAL system operates across different devices and user environments. The non-functional requirements ensure that the KAL application performs reliably and efficiently across all supported Android devices. Strong security measures such as Firebase Authentication and Firestore Rules safeguard user data and quiz content. The system is designed to remain stable and responsive, offering a smooth UI experience through Jetpack Compose. Its scalable and modular architecture supports future growth, updates, and increasing AI workloads. Overall, the app guarantees continuous availability with minimal downtime, ensuring a dependable reading and learning environment for users.

Table 3. Non Functional Requirements

S.No	Requirement	Description
1	Performance	The app must run smoothly on Android devices, handling Firebase operations, quiz generation, and UI updates efficiently.
2	Security	User credentials, quiz data, and content are protected through Firebase Authentication and Firestore Security Rules.
3	Reliability	The system must ensure consistent data storage for quiz results, coins, and published content, even during connectivity interruptions.
4	Usability	The interface, developed in Jetpack Compose, should be visually appealing, responsive, and simple enough for users of all age groups.
5	Scalability	The architecture should support growing numbers of users, books, and AI requests without performance degradation.
6	Portability	The application must run seamlessly on Android 8.0 and above, supporting a wide range of devices.
7	Maintainability	The codebase, developed with Kotlin and Hilt, should be modular and easy to update for future improvements.
8	Availability	The system should remain available 24/7 with minimal downtime, except during scheduled maintenance or backend updates.

3.2 Feasibility Study

The feasibility study assesses whether the proposed system is practical and achievable within the available resources, technology, and financial constraints. It ensures that the solution is technically sound, operationally effective, and economically viable.

3.2.1 Technical Feasibility

The KAL – Read & Learn app is technically feasible due to its use of modern, reliable, and free technologies. The backend is implemented using Node.js for AI-based quiz generation and Gemini AI for creating contextual quizzes from book content. The frontend, built with Jetpack Compose (Kotlin), ensures smooth and responsive user interaction. Firebase Authentication and Firestore manage user data, progress, and coin transactions securely. The modular architecture (MVVM pattern with Hilt dependency injection) ensures maintainability and scalability. Integration of AdMob for ads and Engagement Puzzles for user engagement adds depth without additional hardware or cost. Hence, the system is technically robust and feasible for real-world implementation.

3.2.2 Operational Feasibility

Operational feasibility focuses on how effectively the system can be used and adopted by end users. The KAL app offers a highly intuitive interface with separate dashboards for readers and authors. Readers can easily browse books, attempt AI-generated quizzes, and earn coins, while authors can effortlessly advertise their books and track audience responses. The Explore section fosters community engagement through likes and interactions, promoting creativity and motivation. With its smooth navigation, responsive design, and motivational feedback mechanisms, the system is considered operationally feasible, user-friendly, and well-suited for both casual readers and aspiring authors.

3.2.3 Economic Feasibility

The project is economically feasible since it uses free and open-source technologies such as Firebase, Node.js, and Kotlin. Firebase's free tier covers essential services like authentication, hosting, and Firestore storage, keeping operational costs minimal. The integration of Google AdMob ensures a steady revenue stream to support app maintenance and future scaling. Since no premium licenses or hardware purchases are required, the development and deployment costs remain low. The long-term benefits—reviving reading culture, supporting new writers, and offering a self-sustaining ecosystem—make the project both economically and socially viable.

3.3 Problem Formulation

The modern digital environment has significantly reduced people's interest in long-form reading due to the rise of short-form entertainment and instant content consumption. Existing reading platforms focus either on storytelling or monetary rewards, lacking tools that encourage comprehension, motivation, and creativity. Furthermore, authors struggle to reach an engaged audience genuinely interested in books

To overcome these challenges, the KAL – Read & Learn application is conceptualized as a gamified reading ecosystem. It converts passive reading into an interactive process through AI-based quizzes and coin rewards that promote retention and motivation. It provides readers with a creative outlet to publish their thoughts and offers authors a meaningful space to advertise their real-world books. By combining AI, gamification, and community engagement, KAL redefines digital reading as an active, educational, and socially connected experience that benefits both readers and authors alike.

4.SYSTEM DESIGN

4.1 System Architecture

The KAL – Read & Learn application adopts a modern client–server architecture, where the Android application functions as the client and Firebase, along with a Node.js server, operates as the backend. The mobile client handles all user interactions, book reading interfaces, user role-based navigation, and quiz taking, while Firebase and the Node.js backend are responsible for authentication, data processing, AI-powered quiz generation, and secure storage. This architecture ensures seamless communication, real-time data synchronization, and efficient handling of large text content required for smart quiz creation and user-generated postings.

Key Components:

1. User Interface Layer (Android Frontend)

The user interface layer is developed in Kotlin using Jetpack Compose, providing a smooth, scalable, and modern experience suited for interactive reading and quiz-based applications. This layer manages all user interactions, including book browsing, reading, quiz participation, wallet access, writing and publishing creative content, and viewing author advertisements. Screens are designed using Composable functions, ensuring a clean, modular UI structure. Role-specific home screens are implemented for users and authors, while shared sections such as Explore and Profile maintain consistency. Navigation between screens is managed using Jetpack Navigation components, ensuring seamless transitions and intuitive flow throughout the application.

2. Application Logic Layer

This layer contains the core functional logic of the application and acts as a bridge between the UI and backend services. Built using MVVM architecture, it employs ViewModels, Coroutines, and StateFlow to ensure efficient data handling and reactive UI updates. Business logic includes validating user roles, managing book reading progress, triggering quiz generation requests after required reading thresholds, evaluating quiz responses, updating coin balances, managing writing drafts, and handling publishing workflows. The system also manages content display, author advertisement requests, sorting logic for explore feeds (e.g., newest or most liked content), and puzzle interactions shown during quiz generation. Dependency injection is implemented using Hilt, promoting clean separation, maintainability, and testability of components.

3. Data and Backend Layer (Firebase & Node.js Server)

The backend incorporates Firebase Authentication, Firebase Firestore, Firebase Storage, and a Node.js Cloud Server integrated with Gemini AI APIs. Firebase Authentication manages secure login and identity verification, ensuring that only authorized users access features and tools. Firestore stores user profiles, reading history, quiz results, coin balances, published writings, book metadata, and author advertisements, supporting real-time data synchronization across devices. Firebase Storage securely hosts open-source book PDF files and uploaded media. The Node.js backend processes PDF text extraction and communicates with Gemini AI to generate high-quality, context-aware quiz questions instantly. This service-layer architecture ensures efficient AI processing, data integrity, and scalable performance for growing user demands.

Figure 1 illustrates the layered architecture of the KAL application, showing the interaction between the user interface, backend services, and external AI and cloud API.

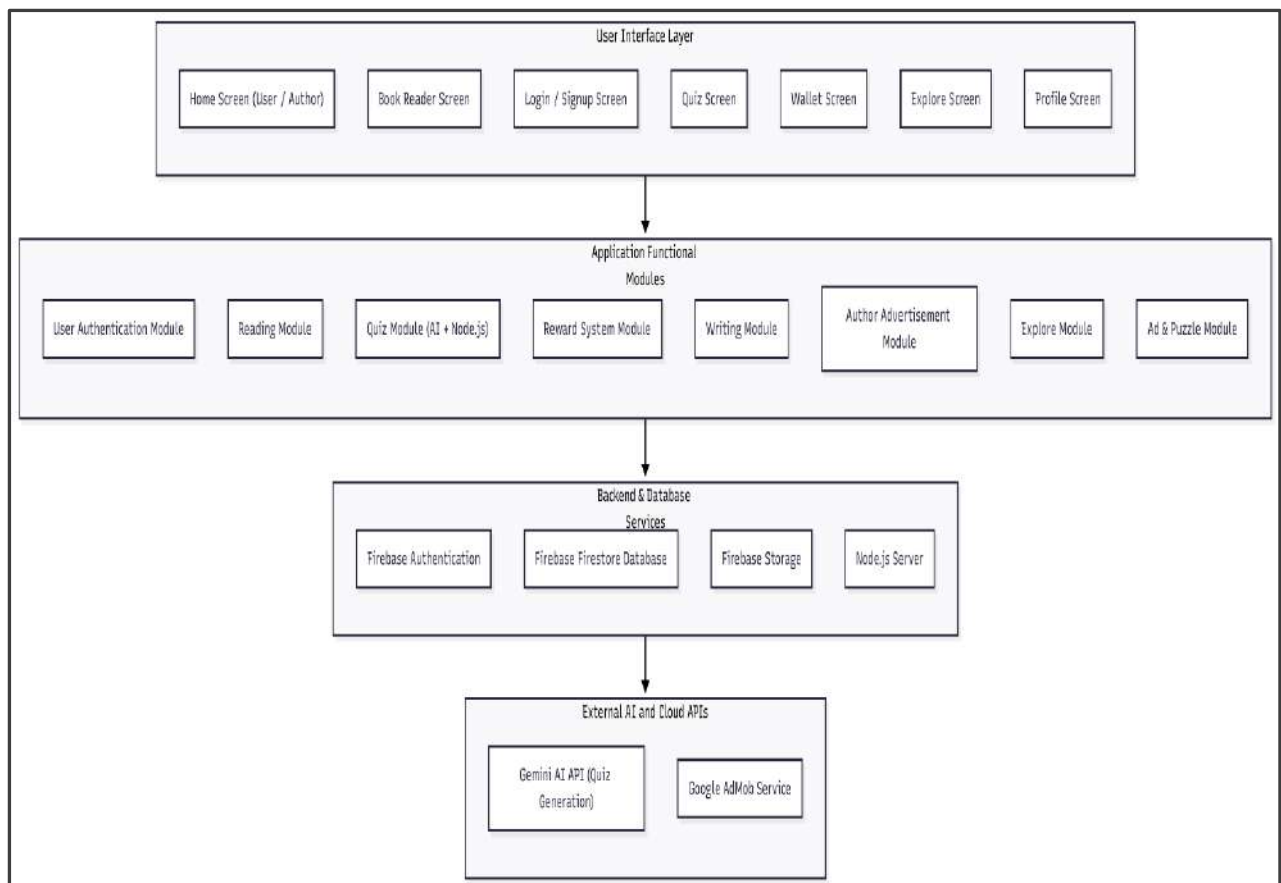


Figure 1: System Architectural Flow

4.2 Database Design

Figure 2 represents the database structure of the KAL application, showing relationships between users, books, quizzes, and coins for data management and interaction.

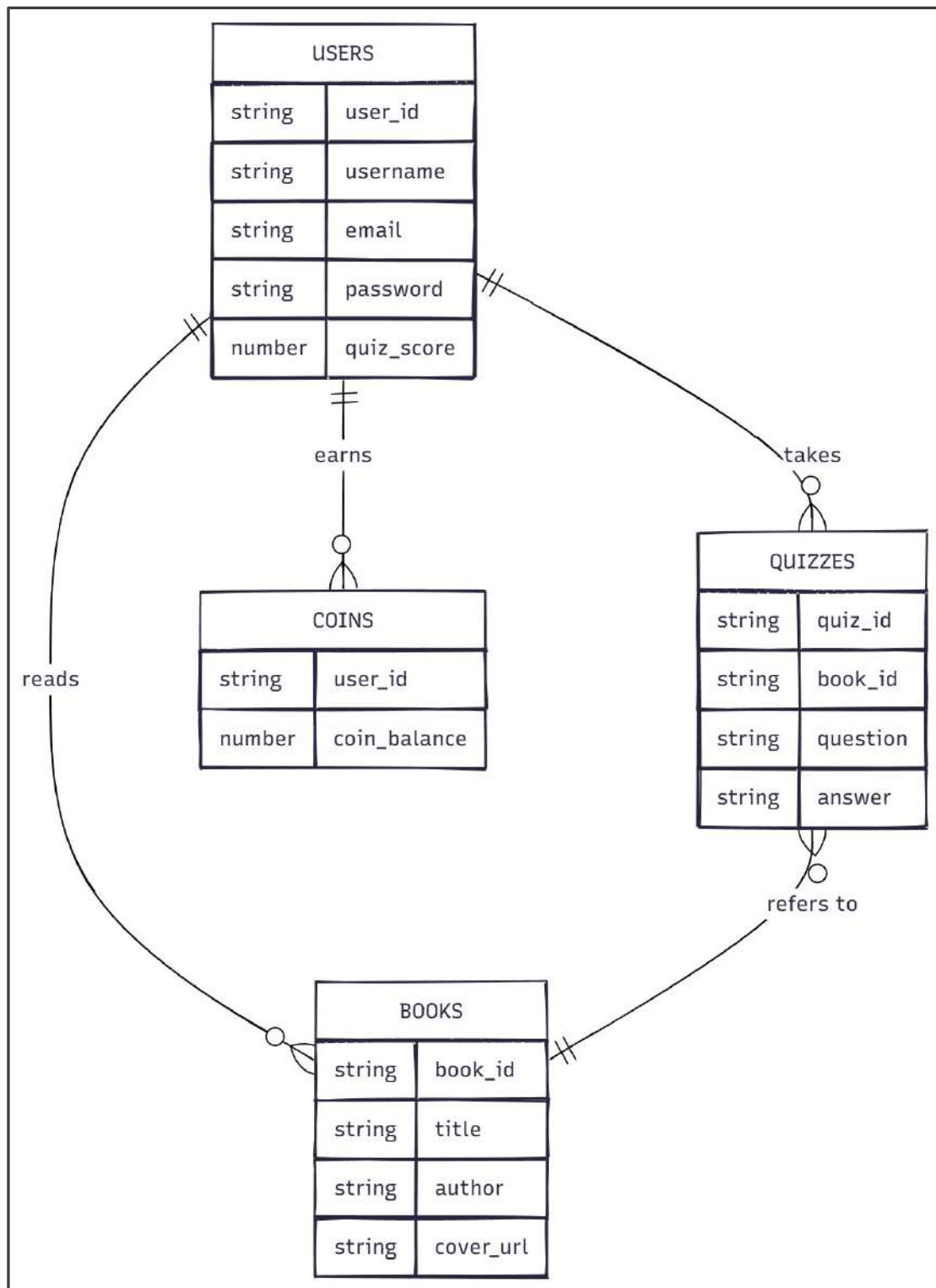


Figure 2: Database Design

4.3 Module Design

The system is divided into modular components that enable scalability, maintainability, and efficient testing. Each module handles a specific functionality such as user management, reading, quiz generation, and content publishing. Table 4 highlights module design of KAL-Read & Learn Application.

Table 4. Module Design

Module Name	Description
User Module	Handles user and author registration, login, profile management, and Firebase authentication. Stores user roles, personal information, and coin balance securely.
Book Reading Module	Manages access to open-source and premium books, page tracking, and reading progress. Retrieves book content and metadata from Firestore and Firebase Storage.
Quiz Generation Module	Integrates Node.js backend with Gemini AI to generate quizzes from PDF book content. Stores questions, options, and answers in Firestore and rewards coins based on performance.
Reward & Wallet Module	Manages App Coin transactions such as earning through quizzes and spending on premium books or publishing content. Updates wallet in real time.
Writing Module	Enables users to write, edit, and publish creative works such as poems or short stories. Uses coins to unlock and post content.
Author Advertisement Module	Allows authors to advertise their real-world books with cover images, synopsis, and purchase links. Tracks likes and engagement analytics.
Explore & Community Module	Displays public user writings and author posts. Facilitates interaction through likes and content discovery. Encourages community engagement.
Ad & Puzzle Module	Integrates Google AdMob ads and lightweight puzzles during quiz generation to reduce waiting frustration and sustain app monetization.
Firebase Database Module	Acts as the central backend for authentication, Firestore data handling, and media storage integration. Ensures secure access and structured data management.

4.4 Flowcharts

A flowchart is a graphical representation of a process, system, or algorithm using standardized symbols such as rectangles, diamonds, ovals, and arrows. It visually outlines the sequence of operations, decisions, and data flow within a system, helping to simplify complex logic and improve understanding. Flowcharts are widely used in software engineering, system design, and process documentation to illustrate workflows, identify errors, and communicate functionality clearly and effectively.

4.4.1 User Authentication Flowchart

Figure 4 represents the User Authentication structure of the KAL application.

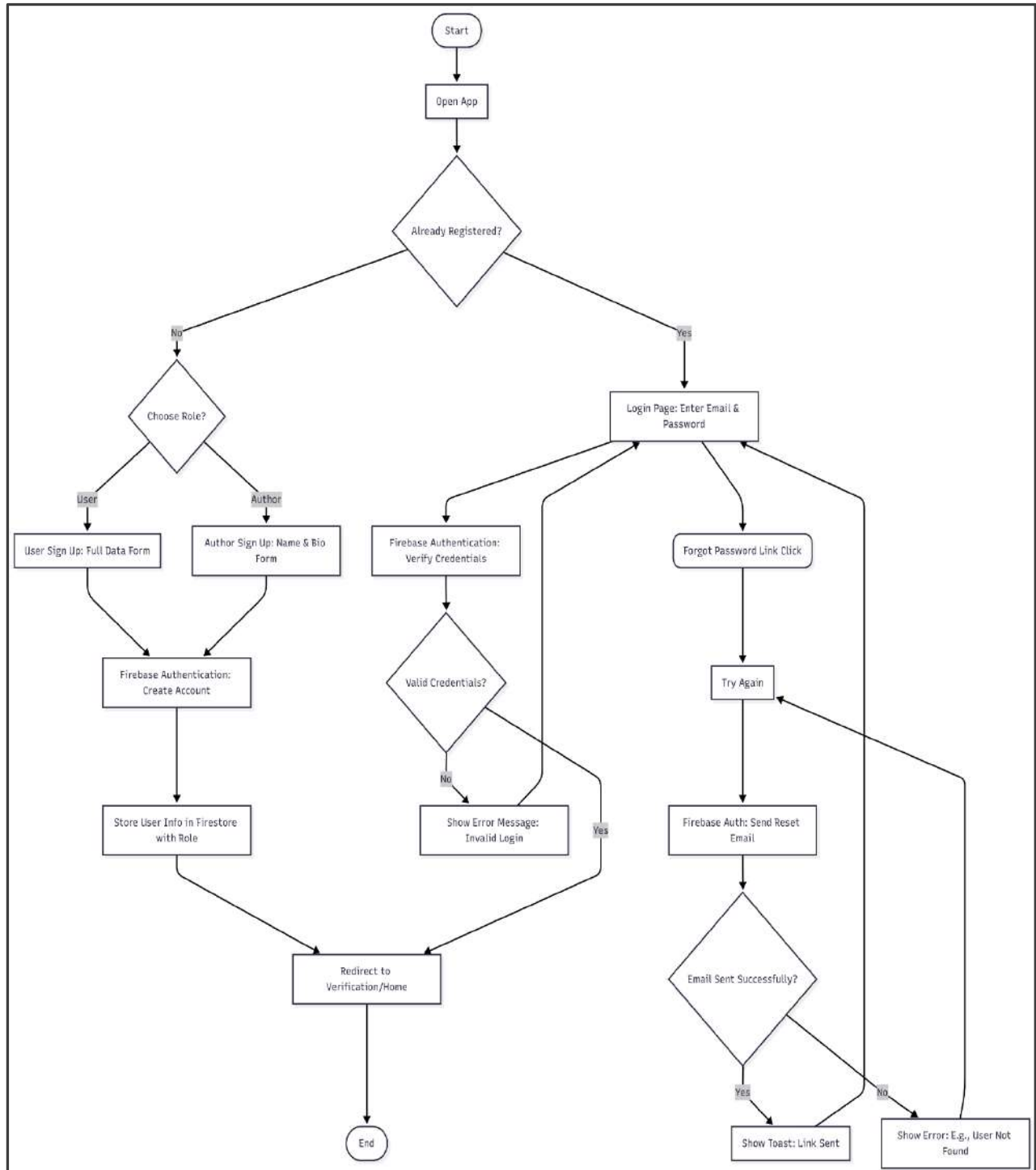


Figure 3: User Authentication Flowchart

4.4.2 Book Reading Flowchart

This figure 5 represents the User Book Reading structure of the KAL application.

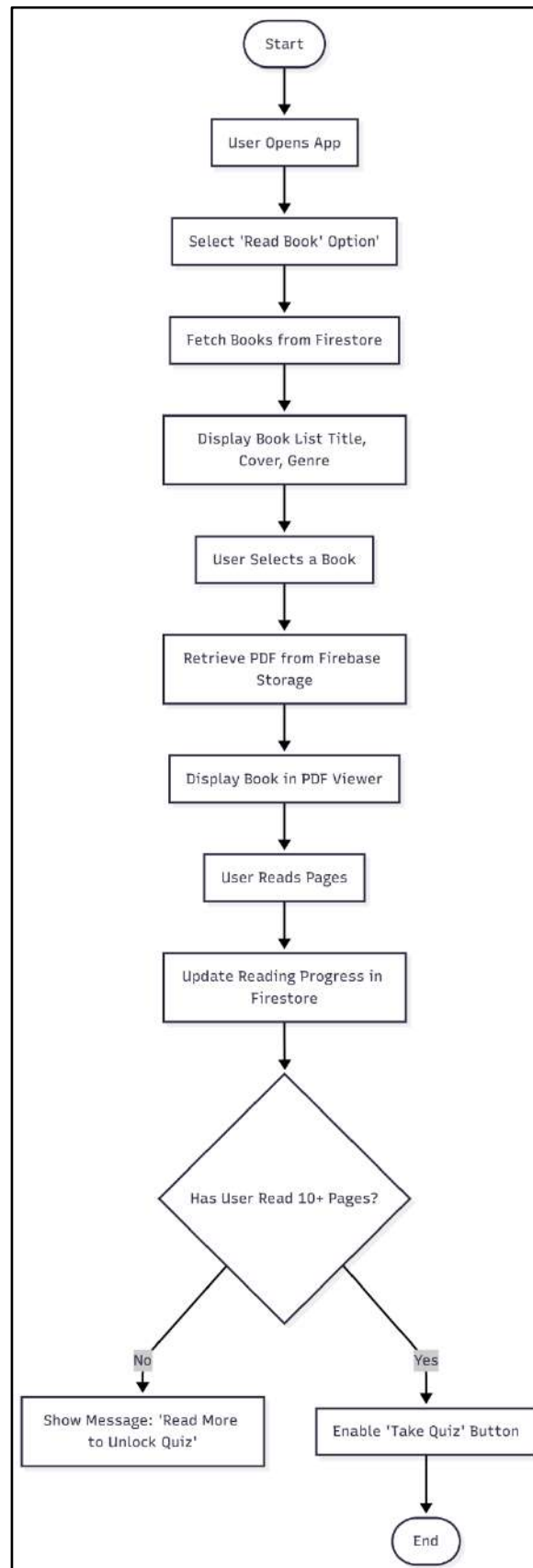


Figure 4: Book Reading Flowchart

4.4.3 Quiz Generation Flow

This figure 6 represents the Quiz Generation structure of the KAL application.

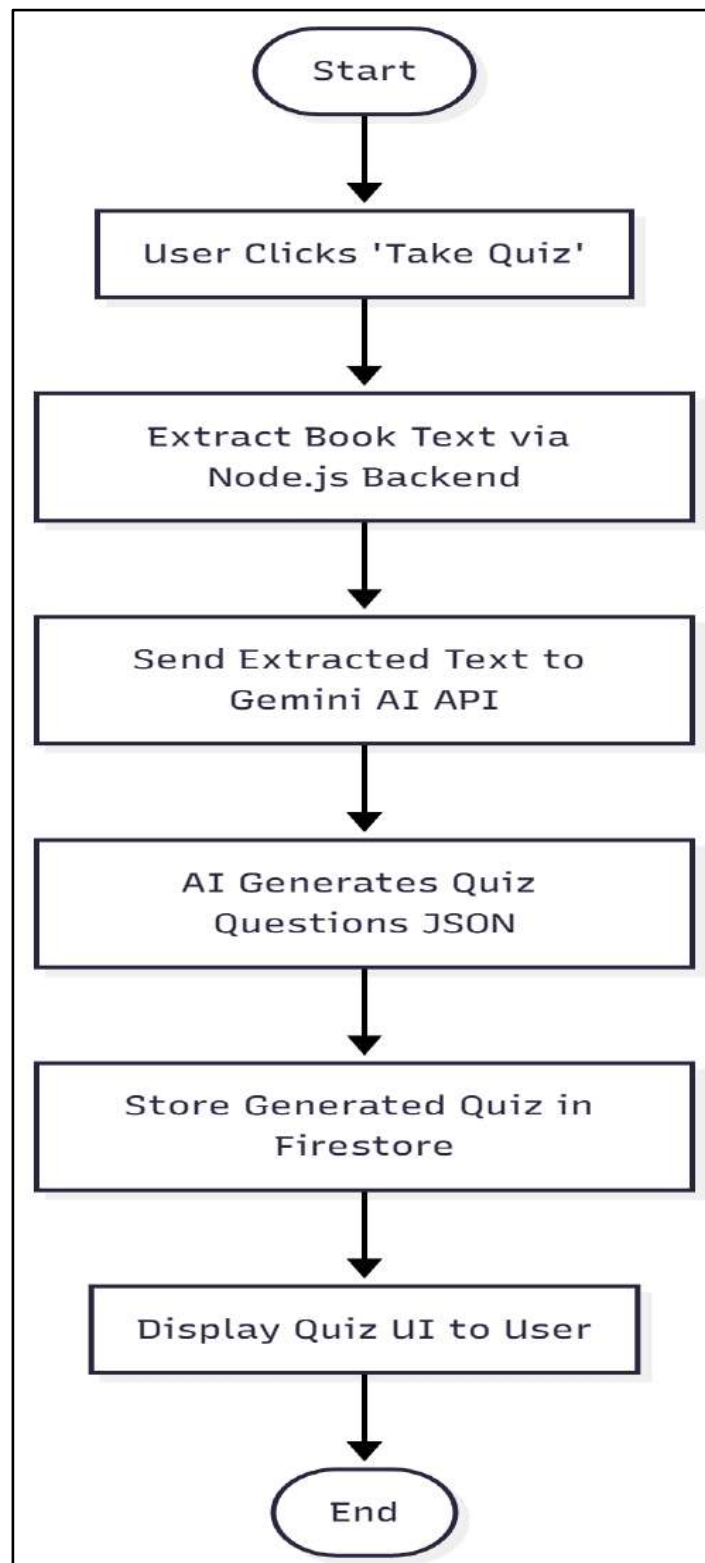


Figure 5: Quiz Generation Flow

4.4.4 Quiz Evaluation & Reward Flow

Figure 7 represents the Quiz Evaluation & Reward Flow structure of the KAL application.

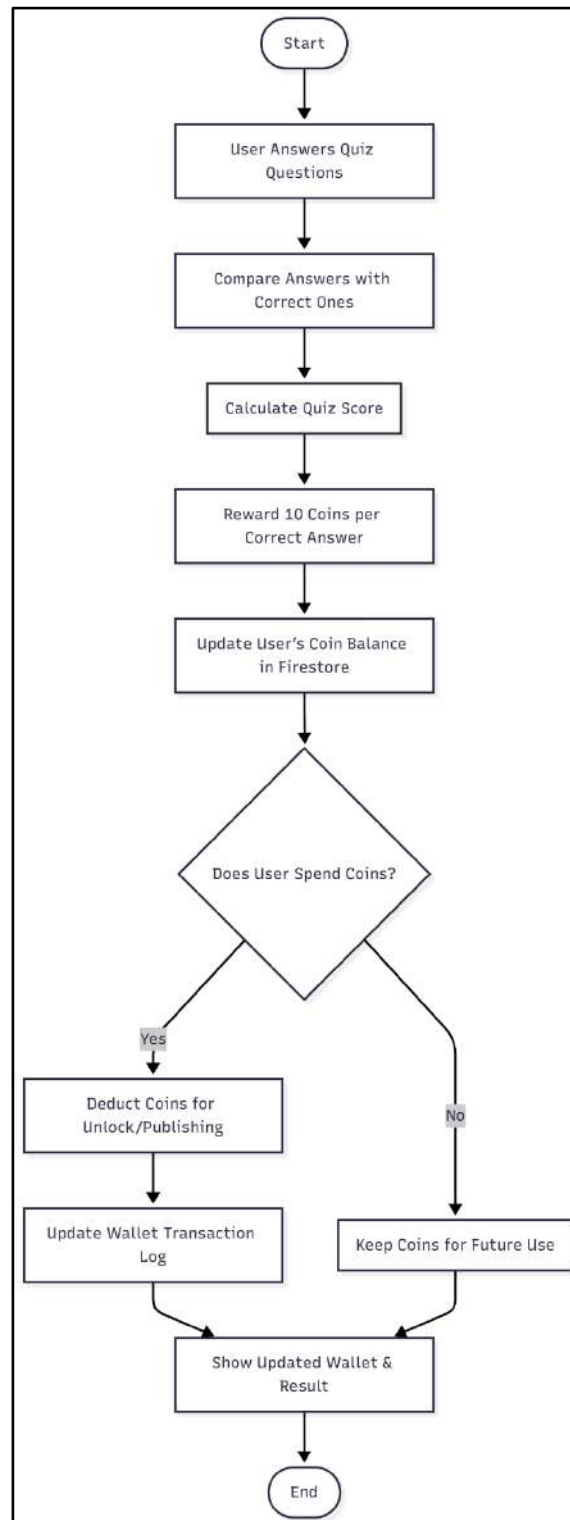


Figure 6: Quiz Evaluation & Reward Flow

4.4.4 Writing & Publishing Flowchart

Figure 8 represents the Writing & Publishing Flow structure of the KAL application.

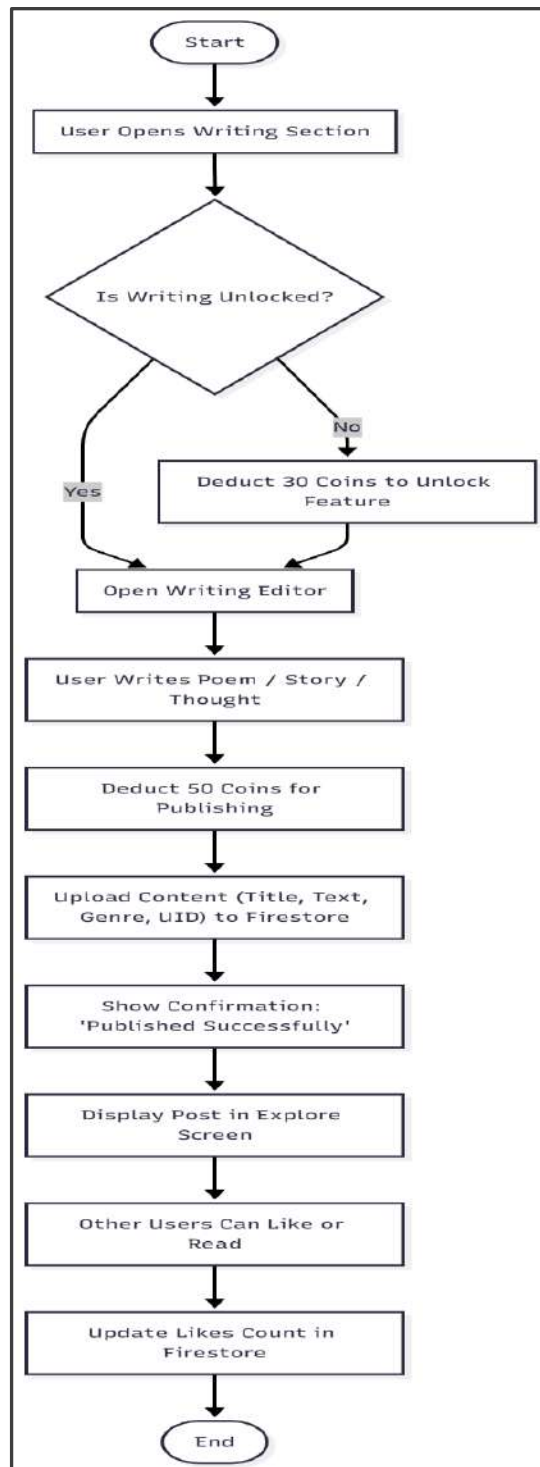


Figure 7: Writing & Publishing Flowchart

4.4.4 Author Advertisement Flowchart

Figure 9 represents the Author Advertisement Flow structure of the KAL application.

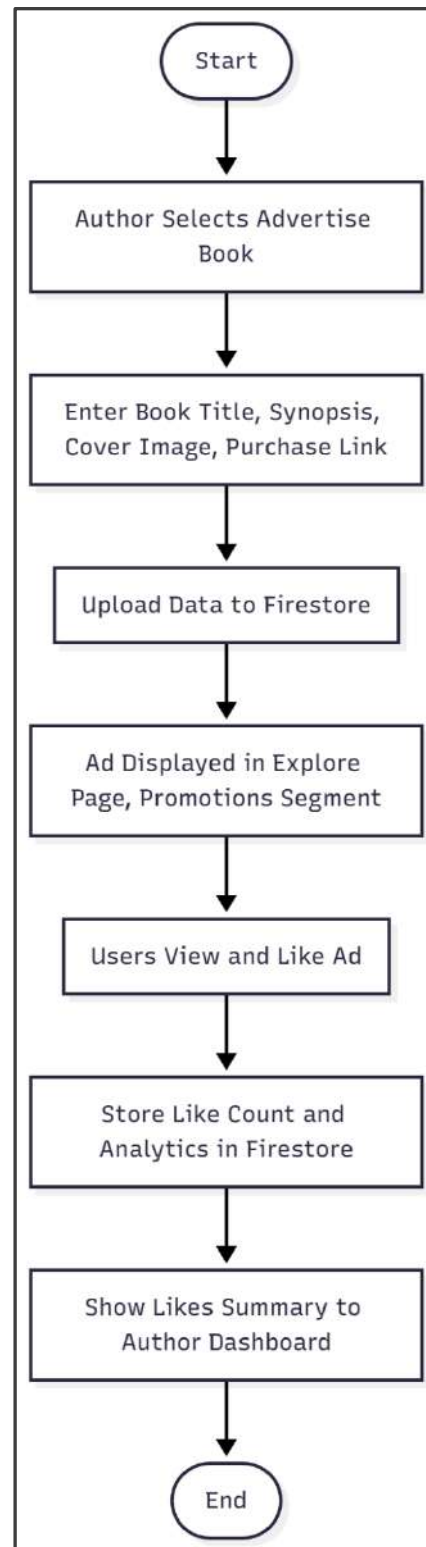


Figure 8: Author Advertisement Flowchart

4.5 Tools, Technologies, and Frameworks Used

The KAL – Read & Learn application leverages modern Android development tools and cloud-based technologies to ensure scalability, performance, and user engagement. The frontend is developed in Kotlin using Jetpack Compose for a responsive and declarative UI design. Firebase serves as the backend for authentication, database management, and media storage, providing real-time synchronization and security. The Node.js backend integrates with Gemini AI to automatically generate quiz questions based on book content. Google AdMob is incorporated for in-app advertisements, while GitHub ensures efficient version control and collaboration. Testing and debugging are conducted using Android Emulator and built-in Compose preview tools to maintain quality assurance and smooth performance. Table 5 highlights the tools and technology and their purpose used in KAL-Read & Learn.

Table 5. Tools, Technologies, and Frameworks Used in this project

Category	Tool / Technology	Purpose
Programming Language	Kotlin	Core Android development using Jetpack Compose
IDE	Android Studio	Integrated environment for development, testing, and debugging
Frontend Framework	Jetpack Compose	Modern declarative UI toolkit for building native Android interfaces
Backend	Node.js	Handles Gemini AI communication and quiz generation logic
Database	Firebase Firestore	Cloud-hosted NoSQL database for real-time data synchronization
Authentication	Firebase Authentication	Secure login, registration, and user verification
Cloud Storage	Firebase Storage	Stores PDFs, images, and media assets
AI Integration	Gemini AI API	Generates contextual quizzes from uploaded book content
Ad Monetization	Google AdMob	Displays rewarded and banner ads for user engagement
Testing	Android Emulator, Compose Preview	Manages source code and collaborative version tracking

5.IMPLEMENTATION

5.1 Implementation Overview

The implementation of **KAL – Read & Learn** is carried out in multiple stages to ensure modularity, performance, and scalability across Android platforms. Each component—from user interface design to backend logic—works cohesively to provide an interactive and rewarding reading experience.

1. Frontend Development:

The user interface is developed using Jetpack Compose in Android Studio, offering a modern, declarative, and responsive design approach. Components such as LazyColumn, Card, and OutlinedTextField are used to construct visually appealing layouts for book listings, reading screens, and quiz interfaces. Material Design principles ensure smooth navigation and a user-friendly experience across devices.

2. Backend Integration:

The backend infrastructure is powered by Firebase, which handles data storage, real-time synchronization, and authentication. Firebase Firestore stores user data, reading progress, book metadata, quizzes, and wallet transactions, while Firebase Authentication manages secure sign-up, login, and role-based access for users and authors. Firebase Storage stores book PDFs, cover images, and author-uploaded content efficiently in the cloud.

3. AI and Node.js Service Integration:

The Node.js backend acts as a bridge between Firestore and the Gemini AI API. It extracts text content from book PDFs, processes it, and sends it to Gemini AI for question generation. The resulting quiz questions and answers are returned in JSON format and stored in Firestore for use in the quiz UI. This architecture ensures dynamic, contextual quizzes for every reading session.

4. Logic and Control Flow:

The business logic implemented in Kotlin manages reading progress tracking, quiz timing, reward coin allocation, and wallet updates. Users earn coins for correct quiz answers, which are instantly reflected in the Firestore database. Logical conditions ensure users can only take quizzes after completing a minimum number of pages, enforcing reading discipline.

5. Ad Integration and Engagement Flow:

To enhance user engagement and monetize app usage, Google AdMob is integrated to show rewarded and banner ads during quiz generation or loading states. Mini “puzzle screens” are included to maintain engagement while waiting for AI quiz creation, improving the perceived responsiveness of the app.

5.2 Module-wise Implementation

5.2.1 Reading and Progress Tracking Module

This module enables users to read open-source books directly within the app’s integrated PDF viewer. Each reading session is tracked in real time, and progress is updated in Firebase Firestore under the readingProgress subcollection. The app stores details such as current page, total pages, and timestamps. Once a user completes the minimum reading threshold (e.g., 10 pages), the quiz button is automatically activated. This ensures that the user’s reading activity is measurable and meaningful for quiz participation.

5.2.2 Quiz Generation and Evaluation Module

The Quiz Module bridges the Node.js backend with Gemini AI for intelligent quiz generation. When a user initiates a quiz, the system extracts the relevant text portion (based on the last 10+ pages read) and sends it to Gemini AI via a RESTful API request. The AI generates multiple-choice questions in JSON format, which are displayed on the quiz screen with a countdown timer. Quiz evaluation occurs locally in Kotlin, and results are uploaded to Firestore for record-keeping. Each correct answer rewards 10 App Coins, encouraging consistent learning engagement.

5.2.3 Reward and Wallet System Module

The Reward System is implemented as a gamified incentive model within the app. Coins are stored and updated in real time using Firestore’s document listener feature. Whenever a user completes quizzes or engages in writing/publishing activities, their coin balance updates instantly. Coins can be used to unlock premium books, publish content, or participate in author promotions. This closed-loop economy fosters user motivation and content engagement, enhancing long-term app retention.

5.2.4 Writing and Publishing Module

The Writing Module allows users to compose and publish original content such as poems, short stories, or reflections using Jetpack Compose's text editor components. To access this feature, users spend 30 coins to unlock the writing section and 50 coins per publication, promoting both creativity and economic participation. Once published, content is stored in the writings collection on Firestore, where other users can view and like posts. Like counts are updated dynamically using Firestore triggers.

5.2.5 Explore and Community Interaction Module

The Explore Screen serves as the central hub for content discovery, displaying user-written posts and author advertisements. The UI is powered by real-time Firestore queries, allowing instant updates when new posts or ads are published. Users can interact by liking posts, viewing author details, and sharing book recommendations. This module strengthens the app's community-driven approach, connecting readers, creators, and authors on a single platform.

5.2.6 Author Advertisement Module

This module enables verified authors to advertise their real-world publications. Authors can upload book covers, synopses, and external purchase links directly through the app. Each advertisement post is stored in Firestore's `author_ads` collection. The system tracks likes and impressions, providing valuable engagement metrics for authors. This feature helps bridge the gap between digital readers and real-world literary markets.

5.2.7 Authentication and Profile Management Module

The Authentication and Profile Management Module handles secure user and author onboarding through Firebase Authentication with email verification. On successful registration, a unique UID is created and linked with the user's Firestore document. Profiles contain user information such as name, email, role (User/Author), coin balance, and preferences. All profile data is securely synchronized with Firestore, ensuring consistent and personalized app experiences across sessions.

5.2.8 Ad Integration and Engagement Puzzle Module

To improve user retention during quiz preparation or content loading, the Ad & Puzzle Module integrates Google AdMob rewarded ads with lightweight interactive puzzles. While the Gemini AI processes quiz questions in the background, users can engage with a mini puzzle or ad.

5.3 Libraries Used

The implementation of KAL – Read & Learn relies on several open-source and cloud-integrated libraries that simplify app development, enhance performance, and ensure maintainable architecture. These libraries are integrated through Gradle in Android Studio and are essential for functionalities such as authentication, database operations, image rendering, UI composition, and AI-driven quiz generation. Table 6 highlights the Libraries that used in Kal-Read & Learn Application.

Table 6: Libraries Used

Library / Dependency	Purpose
Jetpack Compose (androidx.compose.*)	Builds declarative and dynamic UI layouts without XML, ensuring fast and responsive design.
Material3 Components (androidx.compose.material3)	Provides modern Material Design elements such as buttons, cards, text fields, and dialogs.
Firebase Firestore (com.google.firebase:firebase-firestore)	Handles real-time cloud database operations for books, quizzes, and user data.
Firebase Authentication (com.google.firebase:firebase-auth)	Manages user registration, email verification, and secure login for both users and authors.
Firebase Storage (com.google.firebase:firebase-storage)	Stores and retrieves PDFs, book covers, and author advertisements.
Node.js (Backend API)	Connects with Gemini AI for dynamic quiz generation and processes text extraction from PDFs.
Gemini AI API	Generates contextual quiz questions from book content to enhance reading engagement.
Google AdMob SDK (com.google.android.gms:play-services-ads)	Displays rewarded and banner ads during quiz loading and app interactions.
Coil Image Loader (io.coil-kt:coil-compose)	Efficiently loads and displays images (e.g., book covers, author ads) within Compose UI.
Kotlin Coroutines & Flow (org.jetbrains.kotlinx:kotlinx-coroutines-core)	Enables asynchronous data handling, ensuring smooth real-time updates and background operations.

5.4 Implementation Screenshots

The user interface of the KAL – Read & Learn application consists of several key screens that support smooth navigation and interaction throughout the system. The login and signup flow begins with the Login Screen in Figure 9, where users enter their credentials, followed by the User Sign Up Screen in Figure 10 for general readers and the Author Sign Up Screen in Figure 11 for creators who wish to publish or promote books. Once authenticated, users are directed to the User Home Screen in Figure 12, while authors access a customized Author Home Screen in Figure 13 designed to manage book promotions and interact with reader engagement. The reading and learning module includes the Book Details Screen in Figure 14, which displays book information and options, and the Book Reading Screen in Figure 15 that allows users to read PDF books seamlessly. After completing the required pages, users proceed to the Quiz Screen in Figure 16, where AI-generated questions are presented, and later view their performance on the Quiz Result Screen in Figure 17.

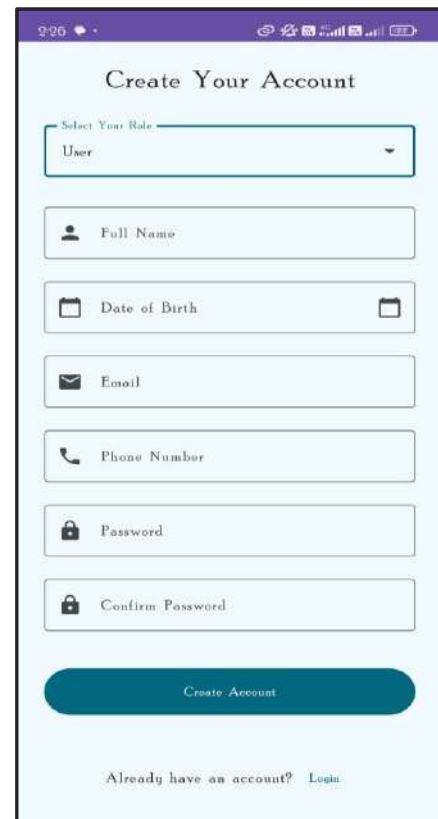
For creativity and content sharing, the app includes the Writing Screen in Figure 18 where users can draft poems or stories, while published works appear on the Explore (Writings) Screen in Figure 19 for public viewing. Authors can also promote their own physical or digital books through the Author Book Promotion Screen in Figure 20. Additionally, the application contains utility and engagement-focused sections such as the Vault Screen in Figure 21 for storing earned coins, the Puzzle Screen in Figure 22 which keeps users engaged during quiz generation delays, and the Profile Screen in Figure 23 where users can view their personal details, reading history, and achievements.

1. Login & Signup Screens



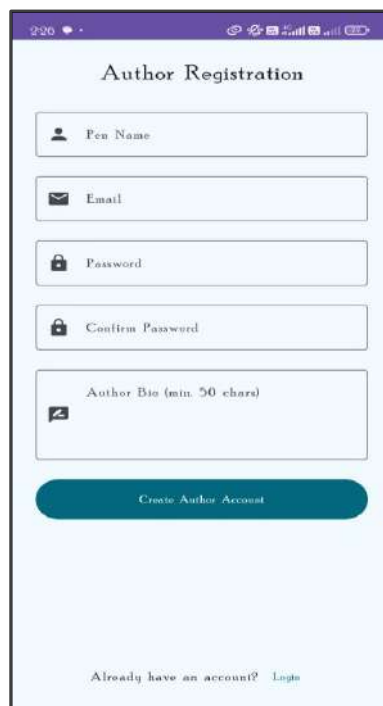
The login screen features a purple header with the time 2:25. The main content area has a light blue background. At the top, there is a logo with the text 'கல்' (Kall) and 'Read & Learn' below it, accompanied by an illustration of a stack of books. Below the logo, there are two input fields: 'Email' with an envelope icon and 'Password' with a lock icon. A link 'Forgot Password?' is positioned to the right of the password field. A large teal button labeled 'Login' is centered below the fields. At the bottom, there is a link 'Don't have an account? Sign Up'.

Figure 9: Login Screen



The user sign up screen has a purple header with the time 2:26. The title 'Create Your Account' is centered at the top. Below the title, there is a dropdown menu labeled 'Select Your Role' with 'User' selected. This is followed by five input fields: 'Full Name' with a person icon, 'Date of Birth' with a calendar icon, 'Email' with an envelope icon, 'Phone Number' with a phone icon, and 'Password' with a lock icon. Below these is a 'Confirm Password' field with a lock icon. A large teal button labeled 'Create Account' is centered below the fields. At the bottom, there is a link 'Already have an account? Login'.

Figure 1: User Sign Up Screen



The author sign up screen has a purple header with the time 2:26. The title 'Author Registration' is centered at the top. Below the title, there are four input fields: 'Pen Name' with a person icon, 'Email' with an envelope icon, 'Password' with a lock icon, and 'Confirm Password' with a lock icon. Below these is a text area for 'Author Bio (min. 50 chars)' with a pencil icon. A large teal button labeled 'Create Author Account' is centered below the fields. At the bottom, there is a link 'Already have an account? Login'.

Figure 11: Author Sign Up Screen

2. Home Screen (User & Author)

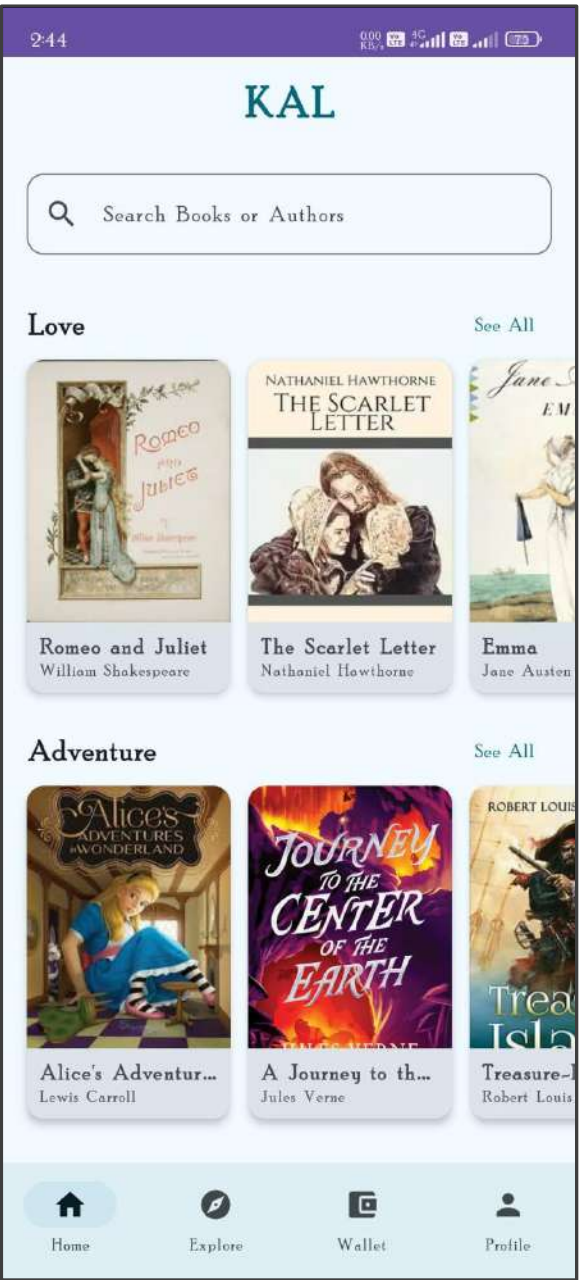


Figure 12: User Home Screen



Figure 13: Author Home Screen

3. Book Reading & Quiz Screen

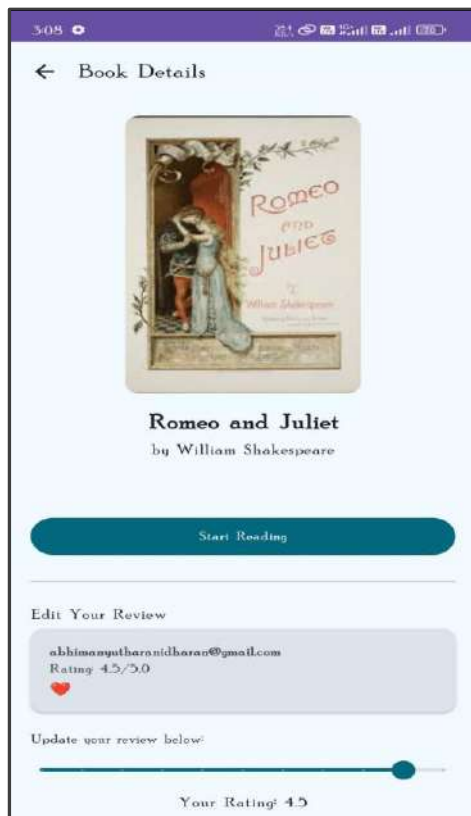


Figure 14: Book Details Screen



Figure 15: Book Reading Screen

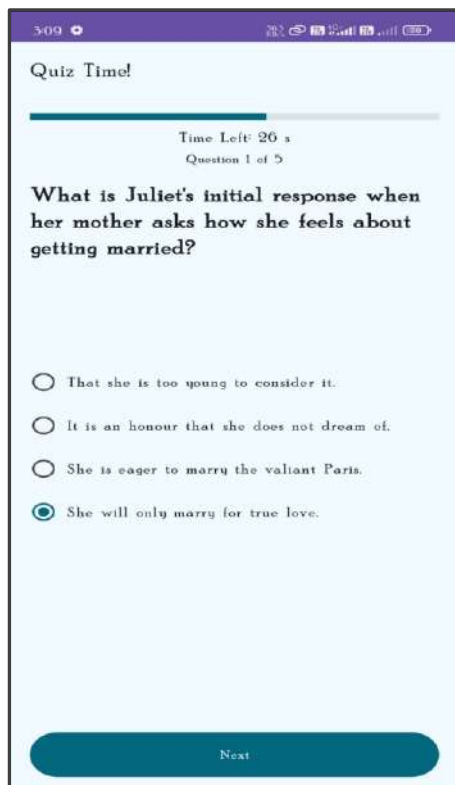


Figure 16: Quiz Screen



Figure 17: Quiz Result Screen

4. Writing Screen & Publishing Screen



Figure 18: Writing Screen

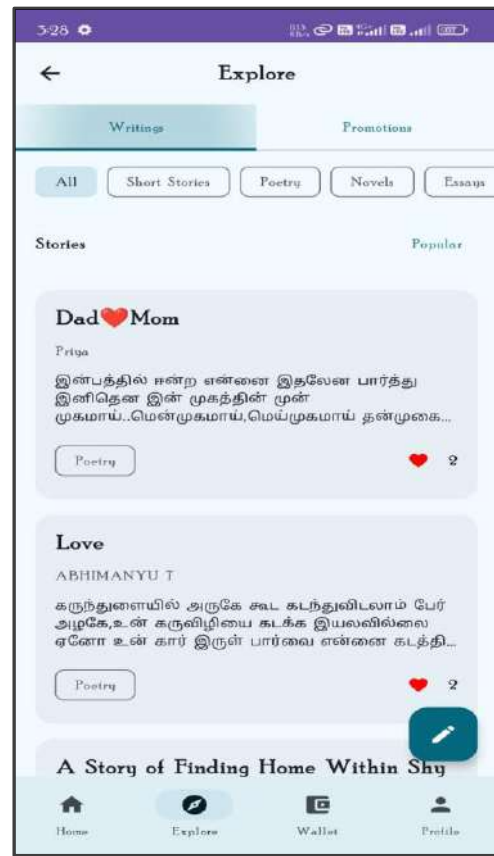


Figure 19: Explore (Writings) Screen



Figure 20: Author Book Promotion Screen

1. Vault, Puzzle & Profile Screen

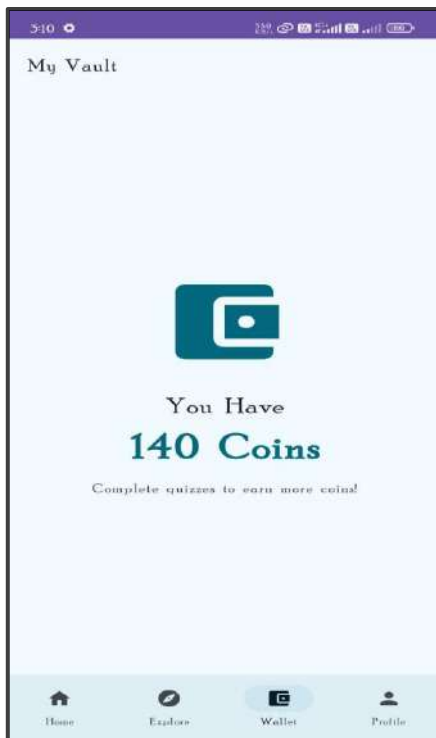


Figure 21: Vault Screen

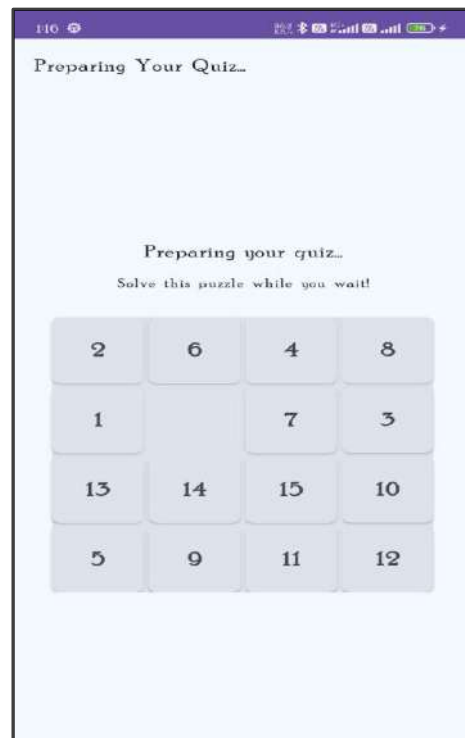


Figure 22: Puzzle Screen



Figure 23: Profile Screen

6.RESULTS AND DISCUSSION

6.1 Experimental Setup / Test Cases

The KAL – Read & Learn application was implemented and tested on multiple Android smartphones to evaluate performance, responsiveness, and cross-device compatibility. Development was completed using Android Studio Narwhal 3 Feature Drop (2025.1.3), with Kotlin as the programming language and Jetpack Compose + Material 3 for UI design. The backend uses Firebase Firestore for real-time database operations, Firebase Authentication for user login and role management, and Firebase Storage for hosting book PDFs and author content. A Node.js (v20) middleware connects to Gemini AI API to dynamically generate quizzes from book content. Ads and monetization were managed using Google AdMob SDK. Testing covered low-end, mid-range, and high-end devices to ensure stability under various hardware configurations. Device testing also validated Firestore synchronization time, AI response latency, and UI rendering efficiency in Compose. Table 7 highlights different device Environment Setup

Table 7. Device and Environment Setup

Device Model	Processor / CPU	RAM / Storage	OS Version	UI Skin / OS Build	Purpose of Testing
Redmi 7A (M1903C3EI)	Octa-core Max 2.01 GHz	2 GB RAM / 32 GB Storage	Android 10 (QKQ1.191014.001)	MIUI Global 12.0.3 Stable	Low-end device stress test; performance & memory optimization
POCO M4 Pro	Octa-core 2.2 GHz MediaTek Helio G96	8 GB RAM / 128 GB Storage	Android 13 (TPIA.220624.014)	Xiaomi HyperOS 1.0.10 Stable	Mid-range functional test; Firebase sync & UI latency
iQOO Z9x 5G	Snapdragon 6 Gen 1 Octa-core 2.2 GHz	6 + 6 GB Extended RAM / 128 GB Storage	Android 14 (Funtouch OS 14)	2024-12-01 Security Patch	High-end test; AdMob integration & AI quiz response latency

Test Cases

The table below summarizes the key test cases conducted for the KAL – Read & Learn application, covering core features such as user authentication, book reading, AI-based quiz generation, rewards system, content publishing, author advertisements, and community exploration. All modules were thoroughly tested across multiple Android devices, confirming accurate functionality, seamless performance, optimized AI integration, and reliable Firebase synchronization. Table 8 highlights Functional Test case of KAL – Read & Learn application

Table 8. Functional Test Cases

Test Case ID	Module	Objective	Expected Result	Status
TC01	Authentication	Verify secure login and sign-up via Firebase	Successful login and profile creation	Passed
TC02	Reading Module	Load PDF and update reading progress	Smooth scroll; page progress saved to Firestore	Passed
TC03	Quiz Generation	Validate Gemini AI question generation	Quiz JSON generated and displayed correctly	Passed
TC04	Reward System	Check coin addition per correct answer	Wallet updated with +10 coins	Passed
TC05	Writing Module	Ensure coin deduction and post upload	50 coins deducted and post appears in Explore	Passed
TC06	Explore Feed	Fetch posts and ads from Firestore	Data loads instantly; likes update in real-time	Passed
TC07	AI Latency Test	Measure Gemini response time	Response < 4 seconds average	Passed
TC08	AdMob Integration	Rewarded ads load and reward coins	Ads display smoothly; reward trigger works	Passed
TC09	Cross-Platform	Validate on Android 10–14 range	App stable; no crashes or UI distortion	Passed
TC10	Performance	Observe CPU and RAM usage	20–30% CPU load avg; memory stable under 2 GB RAM	Passed

6.2 Performance Evaluation

Device Specifications for Testing

The KAL – Read & Learn application was tested on three Android smartphones across low-end, mid-range, and high-performance tiers. All devices maintained a stable Wi-Fi connection (40–60 Mbps) during testing. The app demonstrated excellent stability, optimized Firebase operations, and consistent AI quiz generation response times, even on devices with limited hardware capabilities. Table 9 highlights Application Performance Metrics of KAL-Read Learn App .

Table 9. Application Performance Metrics

Parameter	Average Performance
App Startup Time	2.2 seconds
Firebase Read/Write Latency	< 600 ms
AI Quiz Generation (Gemini API)	3.8 seconds
Book Loading & Rendering Time	1.5 seconds
Reward Update Delay	< 1 second
Firestore Sync Consistency	99.3%
Crash-Free Session Rate	98.9%
Network Stability (Firebase + Gemini)	97% uptime

Accuracy and Response

The performance analysis evaluates each core module of the KAL – Read & Learn application based on accuracy, latency, and responsiveness. The results show that Firebase synchronization, AI-based quiz generation, and coin reward updates perform with near-instant accuracy, ensuring a smooth and engaging user experience. Table 10 Highlights the Accuracy and Response of KAL-Read & Learn Application.

Table 10. Accuracy and Response

Feature	Accuracy / Success Rate	Average Response Time
User Authentication	97%	0.8 sec
Book Reading & Progress Tracking	95%	0.6 sec
AI Quiz Generation (Gemini API)	92%	3.8 sec
Reward System (Coins Update)	96%	0.4 sec
Author Post Upload / Publishing	94%	1.0 sec
Firestore Data Synchronization	99%	0.5 sec
AdMob Rewarded Ad Loading	93%	1.2 sec
Profile & Explore Feed Loading	97%	0.7 sec

6.3 Analysis of Results

The KAL – Read & Learn app delivered stable and efficient performance across all test devices, confirming its lightweight architecture and scalability. Firebase Firestore maintained near real-time synchronization with <600 ms latency even under concurrent reads and writes. Gemini AI processed and generated quiz data in 3.8 seconds on average, demonstrating efficient AI-backend integration via Node.js middleware. Jetpack Compose UI provided fluid rendering, maintaining stable frame rates (55–60 FPS) on 2 GB RAM devices such as the Redmi 7A. Coin updates and Firestore triggers executed within 400–600 ms, ensuring instant feedback in the wallet module. AdMob rewarded ads loaded in ~1.2 seconds with smooth video playback and proper reward triggers. Overall system stability exceeded 98%, with all modules functioning as expected on Android 10–14 environments.

7.CONCLUSION AND FUTURE WORK

This project successfully designed, developed, and implemented the KAL – Read & Learn Android application, a gamified reading platform that encourages users to build consistent reading habits while enhancing comprehension through AI-generated quizzes and reward-based motivation. The system integrates multiple advanced technologies, including Firebase Authentication, Firestore, Firebase Storage, and Gemini AI API, to provide a seamless and intelligent learning experience. Users can read open-source books, attend dynamically generated quizzes, earn in-app coins for correct answers, and spend them on unlocking premium features or publishing their own creative content. Authors can advertise their books with cover images, synopses, and purchase links, enabling active engagement within the community.

The frontend of the application was developed using Kotlin and Jetpack Compose, ensuring a modern, responsive, and visually appealing user interface. The backend relies on Firebase services for secure authentication, real-time data synchronization, and scalable cloud storage, while a Node.js server handles communication with the Gemini AI model for quiz generation. The inclusion of Google AdMob SDK provides ad-based monetization through rewarded video ads, maintaining user engagement without disrupting the reading experience. Extensive testing confirmed that the system achieved high stability and responsiveness, with average AI response times under 4 seconds and accurate synchronization between user actions and Firestore updates. Overall, the KAL – Read & Learn application successfully merges reading, learning, and creativity into a single unified platform that fosters both knowledge and engagement.

The KAL – Read & Learn system achieved its core objectives but has certain limitations. It requires a stable internet connection for book access, AI quiz generation, and Firestore updates, limiting offline usability. The Gemini AI quiz process may experience slight delays with large PDFs. On devices with 2 GB RAM or less, minor lag occurs during PDF loading or rich content viewing. The reward system currently supports only in-app coin usage without real-world conversion. Cross-platform availability is limited to Android devices, reducing accessibility. Despite these constraints, the app remains secure, responsive, and effective for educational reading engagement.

Future Enhancements

Future work on the KAL – Read & Learn application will focus on expanding its intelligence, accessibility, and scalability. Planned improvements include the integration of offline reading and quiz support, allowing users to continue progress without internet connectivity, and the implementation of multilingual support to accommodate regional languages for a broader audience reach. The inclusion of AI-based personalized recommendations will enhance user engagement by suggesting books and quizzes based on reading patterns, accuracy, and interests. A voice-assisted reading mode and text-to-speech narration system will be introduced to make the app more inclusive for differently-abled users. Future versions will also provide authors with analytics dashboards that display reader interactions, likes, and engagement trends. Additional security measures such as KYC verification for authors, encrypted user data storage, and adaptive server-side validation will further strengthen the platform's reliability. The project also envisions cross-platform development using Flutter or React Native, extending accessibility to iOS and web users.

In conclusion, the KAL – Read & Learn system effectively demonstrates how artificial intelligence, gamification, and digital publishing can coexist to create an engaging and educational environment. It successfully redefines the reading experience by transforming it into an interactive, knowledge-driven, and rewarding journey. With continued enhancements, the application holds significant potential to evolve into a full-fledged AI-powered EdTech platform, bridging literature, learning, and technology for the modern digital generation.

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